Network File System

Task 1: Install NFS and create shared folder

On the server VM, run as root:

dnf install nfs-utils -y

mkdir -p /srv/music_share

chmod 755 /srv/music_share

systemctl enable --now nfs-server

On RHEL, the package is nfs-utils. On Ubuntu/Debian, it is nfs-kernel-server.

Task 2: Configure NFS sharing

Find the client IP:

ip addr

Edit the exports file:

nano /etc/exports

Add this line, replacing CLIENT_IP with the client's actual IP:

/srv/music_share CLIENT_IP(rw,sync,no_subtree_check)

Apply the configuration:

exportfs -rav

systemctl restart nfs-server

exportfs -v

If using a second VM, allow NFS through the server firewall:

firewall-cmd --permanent --add-service=nfs

firewall-cmd --reload

Task 3: Mount the share on the client

On the client VM:

dnf install nfs-utils -y

mkdir -p ~/mnt/playlist

mount SERVER_IP:/srv/music_share ~/mnt/playlist

Replace SERVER_IP with the server's IP address.

Test file creation:

touch ~/mnt/playlist/test.txt

ls -l ~/mnt/playlist

cat ~/mnt/playlist/test.txt

If the file creation fails, check directory permissions and whether the client IP matches /etc/exports.

Task 4: Create the restaurant menu folder

On the server VM:

mkdir -p /srv/music_share/restaurant_menus

Copy any two existing PDF or image files into it:

cp /path/to/menu1.pdf /srv/music_share/restaurant_menus/

cp /path/to/menu2.jpg /srv/music_share/restaurant_menus/

Replace the example paths and filenames with your actual files. If you don't have files, you can create or download two sample images/PDFs.

On the client VM, verify:

ls -l ~/mnt/playlist/restaurant_menus

The two files should be visible through the mounted share.

Task 5: Change to read-only access

On the client VM, unmount the share:

umount ~/mnt/playlist

On the server VM, edit the exports file:

nano /etc/exports

Change the line to:

/srv/music_share CLIENT_IP(ro,sync,no_subtree_check)

Apply the change:

exportfs -rav

exportfs -v

Back on the client VM, mount again:

mount SERVER_IP:/srv/music_share ~/mnt/playlist

Test read access:

ls -l ~/mnt/playlist

cat ~/mnt/playlist/test.txt

Test write access:

touch ~/mnt/playlist/newfile.txt

Expected result: The touch command should fail with Read-only file system or Permission denied, confirming that the share is read-only.